

Design Principles for Cold Atom Bitter Coils - DRAFT

Henry Ando,¹ Leon Gold,¹ Geyue Cai,¹ and Cheng Chin¹

The James Franck Institute, Enrico Fermi Institute, and Department of Physics, The University of Chicago, Chicago, Illinois 60637, USA

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We present the design, construction, and characterization of a novel design for Bitter-type electromagnets that can generate large magnetic fields with efficient heat removal for cold-atom experiments. The new design, based on the design presented by Sabulsky et al. 2013, presents an improved and more durable cooling system, and significant reductions in peak thermal resistance, allowing for the safe generation of high magnetic fields with minimal risk of experimental damage from the electromagnets. The electromagnets are constructed from a stack of alternating layers of copper arcs and insulating polyester spacers. Efficient cooling is achieved via perpendicular water cooling channels distributed by a heat-resistant resin block. This ensures that a large portion of the electromagnet is in contact with water cooling, ensuring efficient energy dissipation. Using silicone gaskets and compression, a watertight seal is created, allowing easy disassembly and reassembly of the coil if necessary. At the location of the atoms in our LiCs experiment, 17 mm from the coil's base, the coil has a 1.62 G/A ratio. With 1.28 gal/min cooling water at a temperature of 14°C applied, the coil body heats by 12.2°C under continuous operation at 330 A to a temperature of 28.2°C, with the heat of all components kept below 29°C.

I. INTRODUCTION

Cold-atom experiments require (a) large magnetic fields, (b) the ability to quickly adjust them (low inductance), and (c) thermal stability.

The most common approach to generate these fields is a wound hollow-core wire coil¹⁻³. Recently, other machined geometries have been explored, including Bitter-type electromagnets^{3,4}.

Bitter coils have two key advantages over hollow-core wire coils: (1) The water cooling is in parallel with the current, so the temperature is more uniform across the coil, and (2) the hydrodynamic resistance of a Bitter coil is much lower than that of a hollow-core wire coil, reducing the need for high-pressure pumps and corresponding vibration concerns associated with them.

Previously, we reported on the construction of a bitter coil for cold atom experiments³. After a decade of use, we have identified two primary issues with the original design.

(1) The electrical contact resistance in the bolts that carry the current into and out of the primary coil winding tends to creep upward with time, causing extreme heating, and is then difficult to fix due to access limitations. (2) The plastic water distribution manifold is prone to cracking and leaking.

In this work, we report on a new version of this coil, which seeks to improve on these issues, while also leveraging modern manufacturing technologies such as waterjet cutting, laser cutting, and 3D printing. Our design modifications take inspiration from many other groups^{5,6}, particularly the excellent work by Siegel et al. at NIST⁷.

Our new approach to the current delivery system seeks to reduce heating and improve maintainability by replacing threaded current-carrying connections with flat, accessible surfaces and by water cooling every current-carrying element of the system.

Our new 3D printed water distribution manifold seeks to reduce the likelihood of cracks by reducing the total number of incoming/outgoing water connections from 8 to 2, and by eliminating any threads tapped into plastic, with external tubes mated to compression fittings.

With these improvements, we reduce the maximum temperature increase when producing 1000 G (320 A, 780 W per coil) to under 29 °C, and verify that the water distribution manifold is robust to 2000 full thermal cycles.

II. CURRENT DELIVERY SYSTEM

The main electrical issues we've encountered with the original version of the coil have to do with the connections between the bus bars and the coil stack itself.

In the original design, a current-carrying screw carried current from the bottom layer of the coil to the outgoing bus bar. The full current then passes through the threads of the screw into a copper block. This electrical connection proved problematic over time, as thermal cycling and oxidation gradually caused a runaway increase in contact resistance. Undoing this connection to clean and deoxidize coil surfaces is impossible because the connection in question also provides the clamping force that holds the coil together and the water in.

Another problematic connection was where current was carried from the incoming bus bar into the second current-carrying screw through the threads, then passed into the copper distribution block through the tip of the screw. Both of these connections also developed high contact resistances over time. Since the copper distribution block was recessed, our ability to clean and polish its surface to improve the contact resistance was minimal.

In the updated design, the direction of the compression bolts is reversed: they are installed from the water distribution block downward into blind-tapped holes in the bottom plate. This enables us to separate the mechanical and electrical roles of the current-carrying screw. Now, a nut provides mechanical compression on the coil while the electrical connection is made by a flat surface clamped securely onto the bus bar. Thus, the bus bar can be removed and the connection cleaned, without breaking the water seal. And should the threaded connection between the current-carrying screw and the bottom plate need to be serviced, the blind taps allow the current-carrying screw to be removed through the top of the

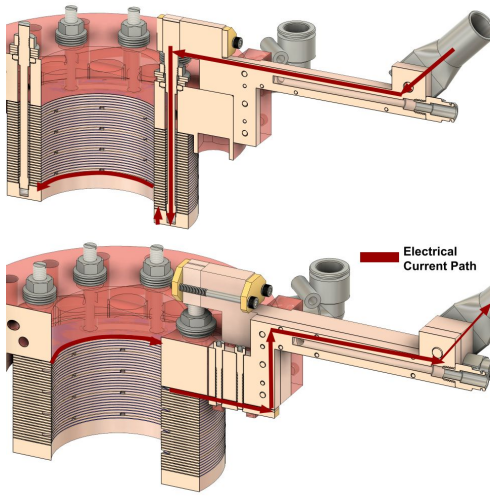


FIG. 1: The current delivery system to the coil, with current brought from a power supply through a water-cooled bus bar, to the base of the coil via its contact with a conducting copper screw, through the coil, and back to ground via the top of the coil's connection with the other water-cooled bus bar.

water distribution block without creating a leak at the bottom of the coil.

Meanwhile, the connection between the incoming bus bar and the top layer of the coil is now made directly, by adding a tongue of metal to the top layer, which is clamped to the bottom of the incoming bus bar. The contact area here is much greater than before, and the bus bar can be removed and all relevant surfaces cleaned/deoxidized if need be, without creating any leaks.

Another general improvement in the new design is that all parts serving both electrical and mechanical purposes (which were previously made of brass) we now make out of chromium-copper C182, inspired by the NIST group⁷. C182 has twice the electrical conductivity of brass while also being stronger and more chemically similar to copper.

The bus bars can get hot, as hot as 40°C when 330 A are applied to the coil, which is why the bus bars are water cooled in parallel with the cooling water going to the coil, although the source is from the same chiller. The bus bars are cooled via internal water cooling cavities installed, connected via a rubber o-ring, and held together via Nylon screws, with the water connectors fitting into NPT taps in these cavities.

III. WATER DISTRIBUTION SYSTEM

The primary issue encountered with the water distribution system in the past was leaks. Leaks mainly occur when the plastic around an NPT-tapped pipe fitting cracks. Once a crack forms, little can be done to remedy the problem in the long term. Common plumbing wisdom advises avoiding plastic/metal pipe connections, especially when a metal male fitting is installed into plastic female threads. This is because plastic is much weaker than metal, especially under tension. Moreover, plastic and metal have different coefficients of thermal expansion, so with thermal cycling, the metal fittings expand and contract relative to their plastic sockets, gradually

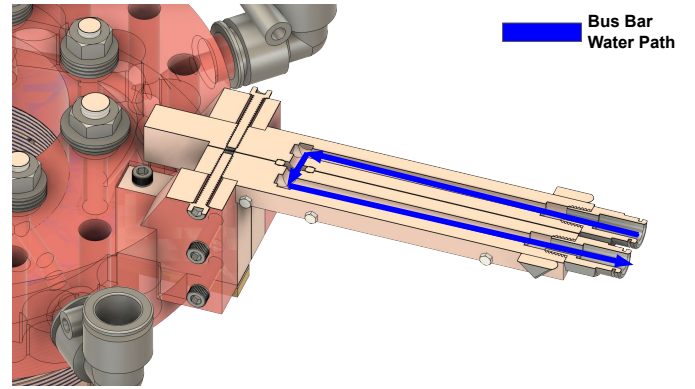


FIG. 2: The water cooling system for the bus bars, that cool the bus bars in parallel with the coil, so that the temperature rise at 330 A remains below 28°C at 330 A. This water cooling system in its present configuration brings 0.47 gal/min through the bus bars.

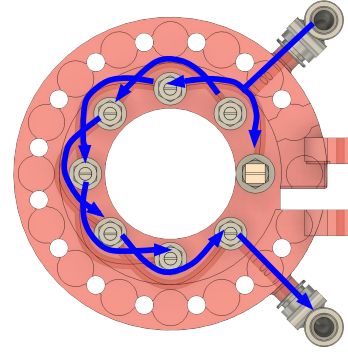


FIG. 3: A top view of the coil's cooling block made out of high temperature resin, with two layers of interior tubing bringing water into and out of the coil through one inlet and one outlet. With our current chiller, 1.28 gal/min can be put through the coil.

stressing the material until it eventually cracks.

The prior design was more prone to leaks with its four water inlets and four outlets, due to geometry limitations imposed by the limits of traditional machining.

In the new design, we leverage the capabilities of 3D printing to address both of the above issues. 3D printing the water distribution block allows us to reduce the number of water inlets and outlets from four each to one each, by routing the water between the vertical channels through internal 3D printed water channels. This technique was also used by the NIST group⁷.

We also remove the need to screw male NPT fittings into female plastic threads by 3D printing an external 0.5" diameter tube, onto which a normal push-to-connect tube fitting can be installed without modification. This eliminates any tensile strain on the plastic piece. Credit for this idea goes to Shaozhen Yang and Rohan Kapur from David DeMille's group⁵.

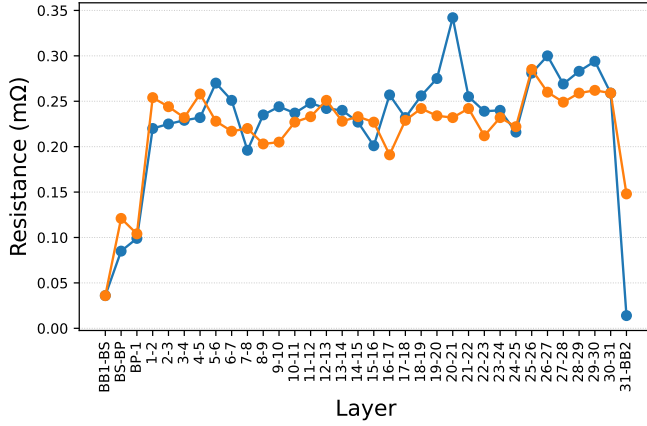


FIG. 4: Electrical resistance between successive coil parts in units of $m\Omega$. The blue line represents the first coil we made (coil 1), and the orange line represents the second coil we made (coil 2). Here, BB1 represents the bus bar connected to the incoming current, BS denotes the conducting screw, and BP refers to the baseplate.

IV. RESULTS

After assembling the latest coil version, we obtain an operational profile by measuring the magnetic field, electrical resistance, inductance, and thermal characteristics during operation.

The electrical resistance of the coil was measured with a voltmeter between the bus bar connected to the incoming current and successive coil layers to obtain a layer resistance profile, while a constant current of 1 amp was applied.

The electrical resistance of the coil can be mathematically estimated from the equation.

$$R = \frac{2\pi N \rho_0}{d \lambda \ln\left(\frac{a_2}{a_1}\right)}$$

where $N = 31$ is the number of turns, $d = 1$ mm is the thickness of a copper arc, $a_1 = 31.75$ mm is the inner radius of the electromagnet, $a_2 = 50.80$ is the outer radius, $\lambda = 0.77$ is the filling factor, and $\rho_0 = 1.7 \times 10^{-8} \Omega m$ is the resistivity of copper. This yields a theoretical resistance of $7.56 m\Omega$.

The theoretical value matches closely the electrical resistance observed of $7.6(1) m\Omega$. This is lower than the resistance in the prior coil by $1.1 m\Omega$.

The reduction in total resistance from $8.71 m\Omega$ down to $7.6(1) m\Omega$ is significant, and this reduction is likely due to a better coil-to-current connection, with these bus bars having minimal contact resistance and better compression techniques. As shown in Fig 4, the layers of the coil account for approximately 95% of the total resistance of the coil, suggesting that much of the resistance attributed to the bus bars from the prior design has been removed and that the new coil is close to the theoretical resistance for this coil design of $7.56 m\Omega$.

To characterize the thermal resistance of the current design, the temperature was measured at 6 points on the coil using thermistors. One sensor was placed on each bus bar, the input and output tubes, the Copper-182 screw, the Copper-182 Plate at the base of the coil, and the Coil Body. The temperature was measured as the current passing through the coil was gradu-

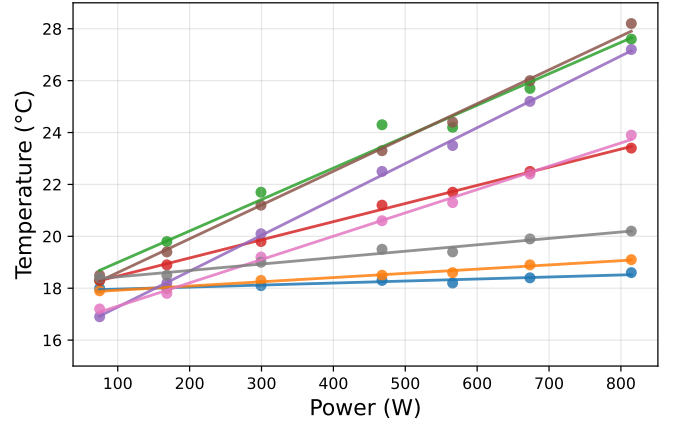


FIG. 5: Temperature of coil components at different power levels. Blue represents the inlet tube, gray the outlet tube, orange the incoming-current bus bar, green the outgoing-current bus bar, red the current-carrying screw, purple the baseplate, brown the lower coil, and pink the upper coil.

ally increased from 0 to 330 amps. During this measurement, the water flow rate was 4.84 L/min into the coil body and 1.8 L/min into the bus bars.

Figure 5 shows the temperature at different coil components as a function of power applied to the coil, with the coil being cooled by 14° water supplied at a constant flow rate of 1.6 gallons per minute. It demonstrates that the maximum temperature change for 20.7°C at 330 A, which is shown in one of the Bus Bars, which heated from 20.3 to 41.2°C . Due to access limitations we don't have a perfectly analogous number for the existing coils installed in our experiment, but a probe on the bus bar of one coil suggests that it regularly reaches 41.2°C running 330 A at a 50% duty cycle with a similar water flow rate.

The input water was measured to have a temperature of approximately 14.0°C , although the tube warmed by 2.4°C as the current increased to 330 A. The output water was warmer by 2.6°C . The lower coil body increased in temperature by 12.7°C from 15.5°C to 24.0°C . The coil baseplate made of Copper-182 increased in temperature by 5.9 degrees from 16.0°C to 21.9°C . These components were the parts of the coil kept the most cool, due to high exposure to the cooling water, as part of the cooling system design. The head of the Copper-182 screw had an initial temperature of 19.1°C , and as the coil increased to 330 A, the temperature rose by 12.2°C to 31.3°C . The bus bars started at respective initial temperatures of 20.5°C and 20.3°C and rose to 41.2°C and 35.5°C after the current was raised to 330 A.

We will briefly describe the magnetic field and inductance results. The magnetic field was tested inside and outside the coil using a gaussmeter, with the bottom of Copper-182 base of the coil used as a reference point. A constant current of 100.01 A was applied to the coil during the testing.

The theoretical field value of the coil at the location of the atoms was calculated to be $1.56(3) \text{ G/A}$. This was predicted from an equation calculating the magnetic field from the coil based on the distance from the bottom of the coil:

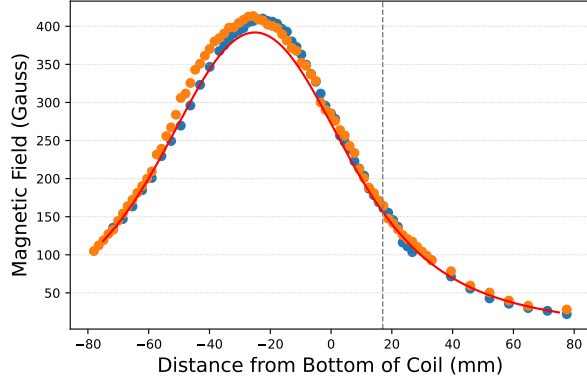


FIG. 6: Magnetic field of coils with 100.01 amps applied as a function from the distance to the bottom of the coil's baseplate. The red line represents the simulation of the field described by the equation below. The blue data points represent the measured field of coil 1, and the orange data points represent the measured field of coil 2. The gray dashed line represents the location of the atoms 17 mm from the bottom of the coil.

$$B_z(z) = C \sum_{k=1}^{k=32} \frac{\mu_0 d \lambda}{2} \cdot \frac{I}{d \lambda \ln(a_2/a_1)} \left(\frac{1}{\sqrt{a_1^2 + (z - z_k)^2}} - \frac{1}{\sqrt{a_2^2 + (z - z_k)^2}} \right)$$
 where C is the correction constant of $7/8$ reflecting that the current only goes $7/8$ of the way around the coil, d is the thickness of an annulus, λ is the copper filling factor, I , a_1 is the inner radius, a_2 is the outer radius, z is the distance from the center of the coil, and z_k is the distance between the annulus and the coil bottom $z_k = z_{face} + (k-1)d$. where z_{face} is the thickness of the base plate.

At 17 mm from the bottom of the coil, the distance between the bottom of the coil and the center of the Li-Cs science chamber, the total measured field had a value of approximately 1.59(3) G/A. The magnetic field was experimentally found by applying 100 A to the coil, and measuring the magnetic field at various distances from the bottom of the coil. A profile of the magnetic fields in both coils and the comparison to the simulated field based on the equation above is shown in figure 3, which shows that the measured field closely matches the simulated field.

The theoretical inductance of the coil can be derived from the equation³

$$L = \frac{\mu_0 N^2}{4\pi} a_1 \Lambda \left(\frac{a_2}{a_1}, \frac{t}{a_1} \right)$$

. Here t is the half height of the coil, and Λ is the self inductance factor, which we found to be 15. This equation yields a theoretical inductance of 45.8 μH .

The impedance of the coil was measured using an oscilloscope from a circuit that applied a variable current to both the coil and a 10 Ω resistor. The impedance was defined as the ratio of amplitudes between the resistor and the coil (divided by 10 to get it in standard units of Ohms). After graphing impedance we were able to estimate the inductance of the coil to be 42.6 μH , which is 4.81 μH less than the measured value from the prior coil. It is also close to the theoretical inductance of 45.8 μH .

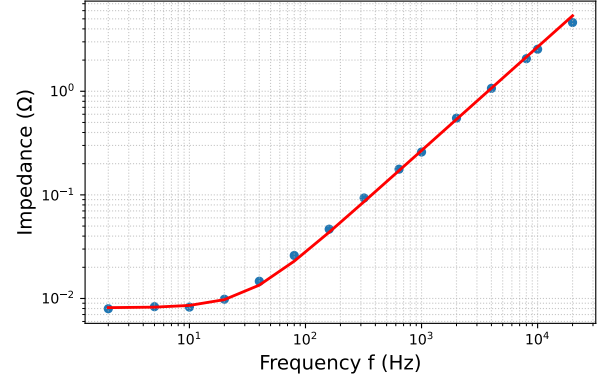


FIG. 7: The impedance of the coil at different applied frequencies, with the fits for resistance and inductance labeled. The data points represent the measured impedances, while the red line is the line of best fit, with the inductance at 42(1) μH and the starting resistance at 8.1(3) m Ω .

V. CONCLUSION

We have described a state of the art, simple, and scalable design of a Bitter electromagnet design for cold-atom experiments alongside other potential uses in industry or quantum computing. During high field operation the coil exhibits stable resistances, temperatures, inductances, and magnetic fields, and outperforms many peer electromagnets, including the prior design published from this group.

The new coil provides improved thermal, electrical, and mechanical performance when compared to our prior coil design and prototypes we tested, with the hottest components under 27 °C, the electrical resistance at 7.6 m Ω , and the inductance under 43 μH during regular operation. Additionally, the coil body and water distribution manifold has reduced tensile stress, which is a significant improvement. The new cooling system eliminates leaks, reduces tension, and has simplified plumbing, with only one inlet and one outlet, improving machinability.

This coil is also very reproducible, with many of the components, including the coil annuli, copper edge pieces, silicone gaskets, and polyester spacers being able to be produced in a standard machine shop with a laser cutter and a CNC or Waterjet. The water distribution manifold, tapped base plate, bus bars, and bus bar connectors needed to be machined externally, but these production costs were in total well in the budget of a standard lab.

Overall, this design details a low cost, state of the art, thermally stable, and machinable electromagnet that should provide stable fields for cold atom experiments.

VI. ACKNOWLEDGEMENTS

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VII. CITATIONS

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